

Many open source tools are available to us but none match *tcpdump*. It is the best tool to capture and filter packets using a basic C code – something that I will discuss briefly, subsequently.

For Windows, there are many tools such as Wireshark and WinPcap. The links are given below.

<http://www.winpcap.org/install/default.htm>

<http://wiki.wireshark.org/Tools>

You can get a list of all the tools in open source from here.

<http://www2.opensourceforensics.org/tools/network>

Everything can be done in the terminal with the help of *tcpdump*. The remaining tools give us easy access, some benefits and some ready-made features.

Capturing one's own data

It's very easy to capture your data in Linux by using the terminal (*Ctrl+Alt+T*) and typing the following command:

```
sudo tcpdump -w capture.pcap
```

This command invokes the tool *tcpdump* and writes the data into the file name 'capture.pcap' (remember .pcap is the extension for the packets captured). This goes on until you give it a keyboard interrupt (*Ctrl+C*) or temporarily stop the job with *Ctrl+Z*. Packet capture can be resumed by giving the command *fg* which is the same as the normal command.

Packet capture can also be done by using online tools such as Wireshark.

Have a look at the man page of *tcpdump* before going ahead with the article.

```
man tcpdump
```

Analysis

I assume that you have a good knowledge of commands (such as *|* and ***) used in the terminal, for this section. If not, here are some links to tutorials that will teach you the basics.

http://linuxcommand.org/lc3_learning_the_shell.php

<http://ss64.com/bash/>

<http://www.pas.rochester.edu/~pavone/particle-www/telescripts/ComputerCommands.htm>

The best way to learn about them is to refer the man page when you have a doubt about a particular command.

Count of packets

To determine the count of packets in the file, we use the following command:

```
tcpdump -nn -r capture.pcap | wc -l
```

Since the file usually contains a large amount of data, instead of using the 'cat' command, it is better to pipe the

```
bios@bios-amrita:~$ sudo apt-get install tcpdump
[sudo] password for bios:
Reading package lists... Done
Building dependency tree
Reading state information... Done
tcpdump is already the newest version.
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
bios@bios-amrita:~$
```

Figure 1: Installing *tcpdump*

```
bios@bios-amrita:~$ sudo tcpdump -w capture.pcap
[sudo] password for bios:
tcpdump: listening on eth0, link-type EN10MB (Ethernet), capture size 65535 bytes
^C10 packets captured
10 packets received by filter
0 packets dropped by kernel
bios@bios-amrita:~$
```

Figure 2: Using *tcpdump* to capture data

file to 'head' so that we can get a clear view of it. This can be achieved by using the following command:

```
tcpdump -nn -r capture.pcap | head
```

The fields of the PCAP file would be in the following order:

1. Time
2. Network protocol
3. Source IP
4. Source Port
5. Destination IP
6. Destination Port

To concentrate on one of the above fields, remove some of them by using commands for piping and filtering. For example, the following command is used to get only the source IP address and its Port:

```
tcpdump -nn -r capture.pcap | cut -f 3 -d " " | head
```

To filter the file to get TCP/IP and exclude the Layer 2 traffic, add the option 'tcp' or 'udp' at the end of the command:

```
tcpdump -nn -r capture.pcap 'tcp' or 'udp' | cut -f 3 -d " " | head
```

To get only the IP address without the Port, just cut the other columns starting from ":

```
tcpdump -nn -r capture.pcap 'tcp' or 'udp' | cut -f 3 -d " " | cut -f 1-4 -d "." | head
```

The 'uniq' command

The 'uniq' command can be used to remove repeated lines

```
bios@bios-amrita:~$ tcpdump -nn -r capture.pcap | wc -l
reading from file capture.pcap, link-type EN10MB (Ethernet)
10
bios@bios-amrita:~$
```

Figure 3: Count of packets